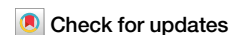


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Improving the performance of lead-free perovskite solar cells

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Lead-halide perovskite solar cells (PSCs) have emerged as promising candidates for next-generation photovoltaics, yet the toxicity of lead poses a significant challenge to their sustainable development. Lead-free analogues, although exhibiting electronic structures and bandgap characteristics distinct from those of Pb-based perovskites, have demonstrated promising optoelectronic properties and improved stability, making them attractive candidates for next-generation photovoltaic applications. This review systematically examines a range of lead-free alternatives, including perovskites derived from isovalent or aliovalent substitution of lead, chalcogenide-based perovskite analogues, and metal-free organic perovskite systems. Recent advances in enhancing power conversion efficiency (PCE) through synergistic material design and device architecture optimization are highlighted. Beyond summarizing recent progress, this review provides a comparative perspective across different lead-free material families by correlating their electronic structures, defect-tolerance characteristics, photovoltaic performance, and sustainability considerations. Such a cross-material analysis reveals the fundamental factors governing device performance, identifies the distinct advantages and limitations of each material class, and provides insights into material-selection strategies for future photovoltaic development. Finally, we discuss the underlying origins of the performance gap between lead-based and lead-free PSCs and offer forward-looking perspectives on the cross-disciplinary approaches needed to overcome existing efficiency and stability bottlenecks.

Photovoltaic technology, renowned for its exceptional sustainability and environmental benignity, has emerged as one of the most promising approaches for harnessing the virtually inexhaustible energy of sunlight^{1,2}. In contrast to the finite nature and substantial ecological impact of fossil fuels, solar energy represents a renewable and clean alternative essential for the long-term transition toward carbon neutrality^{3,4}. According to recent global statistics, renewable energy is estimated to account for approximately 40.9% of electricity generation in 2024, with solar power contributing around 6.9% and exhibiting a steady upward trajectory over previous years. This rapid growth highlights the accelerating maturation of photovoltaic technologies and their expanding industrial ecosystem. Through decades of development, the photovoltaic field has evolved from foundational to advanced solar-cell technologies, with successive generations—distinguished by unique materials, architectures, and performance limits—converging to enable highly efficient solar energy conversion.

Since their introduction into the photovoltaic field in 2009, organic–inorganic hybrid perovskite solar cells (PSCs) have rapidly emerged as a compelling focus for next-generation photovoltaic

technologies⁵, distinguished by their exceptional optoelectronic properties and promise of low-cost fabrication. Within just over a decade, the power conversion efficiency (PCE) of PSCs has risen dramatically from an initial 3.8% to 27.2%⁶, with a certified value of 26.95% reported by the National Renewable Energy Laboratory <https://www.nrel.gov/pv/cell-efficiency>, approaching the performance of conventional monocrystalline silicon solar cells. Such rapid progress is enabled by the exceptional intrinsic properties of perovskite materials, including their high absorption coefficients ($>10^5 \text{ cm}^{-1}$), tunable bandgaps (1.2–2.3 eV), low exciton binding energies, long carrier diffusion lengths ($>175 \mu\text{m}$), and excellent solution processability^{7–9}. The rapid advancement of perovskite photovoltaic technology has been propelled by synergistic innovations in both material design and device engineering. From the earliest liquid-electrolyte architectures to solid-state hole-transport-layer devices, and subsequently to inverted configurations and two-dimensional perovskite heterojunctions, researchers have achieved continuous breakthroughs in film fabrication, bandgap engineering, and interface passivation^{10,11}. At present, inorganic and organic–inorganic hybrid perovskites such as CsPbI_3 , FAPbI_3 , and mixed-halide systems (e.g., $\text{CsPb}(\text{Br}_x\text{I}_{1-x})_3$) are widely incorporated into high-

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efficiency devices, demonstrating strong potential for applications in flexible photovoltaics, tandem solar cells, and photodetectors^{12–17}.

To date, lead-based perovskites remain the foremost in both performance and research focus by a wide margin. On one hand, Pb^{2+} possesses an ideal ionic radius and a stable $6s^2$ lone-pair electron configuration, enabling the PbX_6^{4-} octahedra formed with halides to exhibit favorable lattice symmetry and manageable structural distortion¹⁸. This electronic configuration results in a band structure in which the valence-band maximum (VBM) mainly originates from the antibonding interaction between Pb 6s and halide p orbitals (e.g., I 5p), while the conduction band minimum (CBM) is primarily derived from Pb 6p orbitals, leading to a direct bandgap that facilitates strong light absorption and efficient carrier transport^{19,20}. On the other hand, the bandgap of lead-based perovskites can be systematically tuned through the A-site cations (such as CH_3NH_3^+ (MA^+), $\text{CH}(\text{NH}_2)^+$ (FA^+), Cs^+) or halide components (I^- , Br^- , Cl^-), spanning the visible to near-infrared spectrum and meeting the requirements of diverse photovoltaic and optoelectronic applications^{21–24}. Moreover, the relatively high Pb–X bond energies enables the crystals to achieve good crystallinity and robust optoelectronic performance even under low-temperature solution-processing conditions²⁵. Leveraging these advantages, lead-based perovskites have not only achieved record-high efficiencies in photovoltaics but also exhibit outstanding performance in light-emitting diodes (PeLEDs), photodetectors, lasers, and other optoelectronic devices. Consequently, the pivotal role of Pb^{2+} in governing electronic structure, defect tolerance, and photovoltaic performance remains challenging to fully replicate in alternative systems. Although numerous reviews have summarized the progress of lead-free perovskites, many of them have primarily focused on specific material categories, device performance enhancement strategies, or recent efficiency milestones. However, a systematic comparison of different lead-free material families from the perspectives of electronic structures, defect tolerance, photovoltaic applicability, and sustainability remains relatively limited.

In this context, the development of lead-free perovskites and perovskite-inspired materials has attracted increasing attention as a promising route to mitigate toxicity and improve environmental compatibility. It is important to emphasize that, due to the replacement of Pb^{2+} with alternative elements or ions, lead-free systems exhibit fundamentally different electronic structures, orbital characteristics, and bandgap features compared to their lead-based counterparts. As such, their optoelectronic properties should be evaluated based on their intrinsic characteristics rather than simply benchmarked against Pb-based perovskites. Nevertheless, rational compositional design and structural engineering have enabled these materials to exhibit promising potential for optoelectronic applications, particularly in terms of stability and sustainability, although further optimization is still required. Compared with existing reviews that mainly summarize individual lead-free perovskite families, recent device-efficiency records, or specific subclasses such as Sn-based perovskites and halide double perovskites, this review provides a broader comparative perspective by correlating electronic structures, defect characteristics, photovoltaic performance, and sustainability considerations across different lead-free material systems. Rather than treating Pb substitution as a simple chemical replacement strategy, we emphasize that different material families possess fundamentally distinct orbital configurations, lattice characteristics, defect landscapes, and degradation mechanisms, which ultimately determine their photovoltaic feasibility. Through this comparative analysis, several important insights emerge. First, achieving Pb-free photovoltaic performance comparable to Pb-based perovskites requires not only suitable bandgap engineering but also the simultaneous optimization of band-edge orbital alignment, defect tolerance, and carrier-transport properties. Second, different lead-free systems exhibit intrinsic advantages and limitations that define their most suitable application scenarios. For example, Bi/Sb-based low-dimensional perovskite-inspired compounds and halide double perovskites generally exhibit enhanced chemical stability but often suffer from unfavorable band structures, strong carrier localization, or limited carrier transport. Chalcogenide perovskite analogues and metal-free organic

perovskites provide additional opportunities for sustainability-oriented applications but require further improvements in processing and photovoltaic performance. Therefore, the future development of lead-free perovskites should not rely solely on pursuing record efficiencies, but should instead consider the intrinsic material properties and application-specific requirements, including outdoor photovoltaics, indoor photovoltaics, tandem architectures, photodetection, light emission, and other emerging optoelectronic applications.

Despite the superior performance of lead-based perovskites, their practical deployment is still challenged by concerns regarding the environmental and health risks associated with Pb, a well-known toxic heavy metal. Notably, in perovskite photovoltaic devices, these risks primarily arise during material synthesis, device damage, environmental exposure in the absence of effective encapsulation, and end-of-life disposal, rather than during normal operation. Under moisture, oxygen, light, or acidic conditions—particularly when devices are damaged or insufficiently encapsulated—degradation processes may lead to the formation of water-soluble lead species (e.g., PbI_2) that can leach into the surrounding environment. Once released, these species can migrate through water systems and soil, enter the food chain, and accumulate in organisms, thereby posing environmental and health risks, as schematically illustrated in Fig. 1. Fortunately, recent studies suggest that properly encapsulated perovskite devices exhibit minimal lead leakage under normal operating conditions^{26,27}. When accumulated in the environment over time and biomagnified through the food chain, Pb can cause irreversible harm to the nervous, hematological, and reproductive systems^{28–31}. Therefore, beyond achieving high efficiency, addressing lead toxicity requires not only the development of lead-free alternatives but also advances in encapsulation strategies, device stability, and end-of-life management. In particular, low-toxicity lead-free perovskites may offer unique opportunities in emerging application scenarios such as indoor photovoltaics, where material safety, operational stability, and spectral compatibility can be as important as achieving high efficiency.

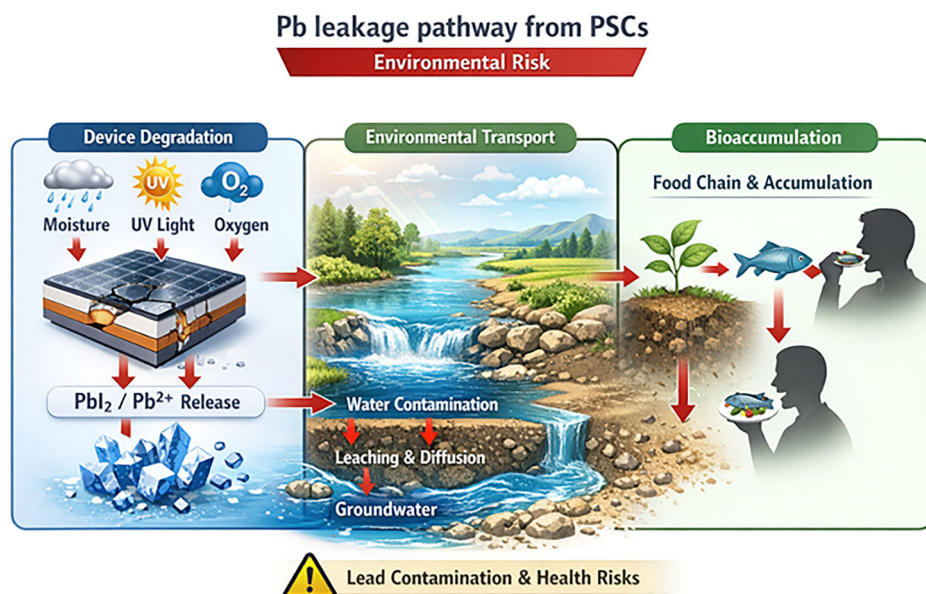
Design principles of lead-free PSCs

The design principles of lead-free PSCs aim to remove toxic lead while maintaining or even improving the optoelectronic performance and stability of the devices. Achieving this goal requires a systematic design strategy across multiple aspects, including crystal structure, band engineering, carrier dynamics, and environmental compatibility. Firstly, the ideal absorber material should preserve the stable ABX_3 perovskite-type framework, in which the ionic radii of the A-site cation (organic or inorganic) and the B-site metal cation satisfy the empirical Goldschmidt tolerance and octahedral factor, thereby ensuring lattice geometry stability^{24,32}. The selection of the B-site metal cation is crucial, as it should possess an electronic structure analogous to Pb^{2+} to achieve comparable band delocalization and efficient light absorption. Commonly explored alternatives to replace Pb^{2+} include Sn^{2+} , Ge^{2+} , Bi^{3+} , and Sb^{3+} . Among them, the Sn^{2+} -based system is considered one of the most promising candidates, as its ns^2 electronic configuration closely resembles that of Pb^{2+} , resulting in high carrier mobilities and a small effective mass^{33–35}.

Band structure engineering is central to determining the performance of lead-free perovskite solar cells. The optimal bandgap should be in the range of 1.3–1.5 eV to optimize the trade-off between light absorption and voltage output, in accordance with the Shockley–Queisser theoretical limit. By tuning the type of B-site cations and the halide composition ratio, continuous and tunable light absorption can be achieved from the near-infrared to the visible region, while ensuring that the valence and conduction-band edges are well-aligned with the energy levels of charge transport layers, thereby reducing interfacial barriers and nonradiative recombination losses^{36–38}. To address this, design strategies must include precursor stoichiometry control, dopant regulation, and surface passivation to suppress oxidation, along with the incorporation of antioxidant ligands or reducing agents to stabilize the valence state of the material^{39–41}.

Stability engineering is essential for the long-term operation of lead-free PSCs. Exposure to heat, humidity, oxygen, and light can induce lattice

Fig. 1 | Lead pollution pathways. Schematic illustration of the migration of lead-based PSCs into aquatic environments.



distortions and ion migration, particularly in Sn-based perovskites, which are highly susceptible to oxidation and structural instability, ultimately leading to device degradation. Incorporation of inert cations (e.g., Cs^+ or FA^+) to enhance crystal symmetry, the use of low- or quasi-two-dimensional structures to suppress ion migration, and surface passivation strategies such as sulfidation, fluorination, or organic ligands can collectively enhance the environmental stability of lead-free perovskites^{42–54}. Simultaneously, interface-layer engineering is equally important, as band alignment and interfacial defect control not only affect carrier extraction efficiency but also determine operational stability under cyclic illumination^{55–57}.

In terms of carrier transport, an ideal lead-free perovskite should exhibit high carrier mobility, long diffusion lengths, and low trap-state densities to ensure efficient separation and collection of photogenerated electrons and holes. Sn-based perovskites exhibit pronounced antibonding features in their electronic structure, which facilitates band delocalization and the formation of high carrier mobilities⁵⁸. However, the inherent oxidative instability of these materials degrades carrier transport through the creation of scattering centers and trapping states. Strategies such as fabricating highly crystalline films, optimizing solvent-processing technique, and employing graded bandgap designs can effectively reduce energy losses and reduce energy loss and facilitate continuous carrier transport across the film. Moreover, interface stress management and the chemical compatibility at the perovskite/electrode interface also affect charge transport efficiency and operational stability^{59–61}, representing crucial directions for achieving high-performance devices.

In summary, the ideal design of lead-free perovskite solar cells should achieve a balanced optimization of structural stability, bandgap engineering, carrier transport, and environmental sustainability. Besides, future design strategies should emphasize sustainable practices across the entire lifecycle, avoiding toxic solvents and high-energy processes from material synthesis to device recycling, while employing degradable ligands and solution-based techniques to achieve low-carbon fabrication. Through the synergistic optimization of chemical doping, interface engineering, and multi-dimensional structural design, it is possible to realize a new generation of perovskite photovoltaic devices that combine high PCE with long-term stability and environmental sustainability.

Potential candidates for lead-free PSCs

Isovalent substitution of Pb

The most straightforward design strategy for lead-free perovskites involves the isovalent substitution of Pb^{2+} with Group IV cations (such as Sn^{2+} or Ge^{2+}), aiming to preserve the original perovskite crystal and electronic

structures, thereby enabling the retention of excellent optoelectronic properties as much as possible. Both Sn^{2+} and Ge^{2+} possess ns^2 electronic configuration similar to Pb^{2+} . Their $\text{ns}^2\text{-np}^0$ antibonding states contribute strong orbital delocalization at the valence-band maximum, resulting in low carrier effective mass and high mobility^{62–65}.

Sn-based PSCs constitute the earliest and most extensively investigated class of lead-free perovskite solar cells. As shown in Fig. 2, perovskite solar cells have experienced rapid gains in power conversion efficiency over the past decade. Within this landscape, Sn-based perovskites represent a key lead-free branch, with reported efficiencies approaching ~18%. However, a clear performance gap remains compared to Pb-based perovskites. This gap originates from two fundamentally different mechanisms: (i) extrinsic chemical instability associated with the oxidation of Sn^{2+} to Sn^{4+} , and (ii) intrinsic defect physics arising from the formation of Sn vacancies (V_{Sn}), which induce strong p-type self-doping. These mechanisms differ in both physical origin and their impact on device performance, and therefore must be addressed separately in material and device design. Sn-based perovskites, such as CsSnI_3 , MASnI_3 , and FASnI_3 , possess an ideal bandgap range of 1.3–1.5 eV, approaching the theoretical optimum for single-junction solar cells. Sn-based PSCs represent the first and most prominent category of lead-free perovskite photovoltaics. In 2012, CsSnI_3 -based PSCs were first reported with a PCE of 0.9%⁶⁶. In practice, these two effects often coexist and synergistically limit device performance, highlighting the need to decouple their contributions through targeted chemical stabilization and defect engineering strategies. Beyond these intrinsic limitations, the degradation of Sn-based perovskites involves complex chemical and structural evolution. Under ambient conditions, exposure to oxygen and moisture induces the oxidation of Sn^{2+} to Sn^{4+} , accompanied by secondary phase formation and increased trap densities. Meanwhile, ion migration and lattice instability accelerate phase degradation and interfacial deterioration. Together, these processes drive rapid performance decay, particularly under continuous illumination and thermal stress, rendering stability the key bottleneck for practical applications. Building on this understanding, diverse strategies have been developed to simultaneously suppress Sn^{2+} oxidation and control defect formation in Sn-based perovskites. Beyond additive engineering, increasing attention has been directed toward the development of DMSO-free processing routes for Sn-based perovskites. Conventional Sn-based precursor solutions typically rely on DMSO, which strongly coordinates with Sn^{2+} and modulates crystallization kinetics, but can also introduce residual solvent complexes and promote oxidation under certain conditions. DMSO-free strategies, such as the use of alternative coordinating solvents and solvent engineering, have been shown to improve film quality,

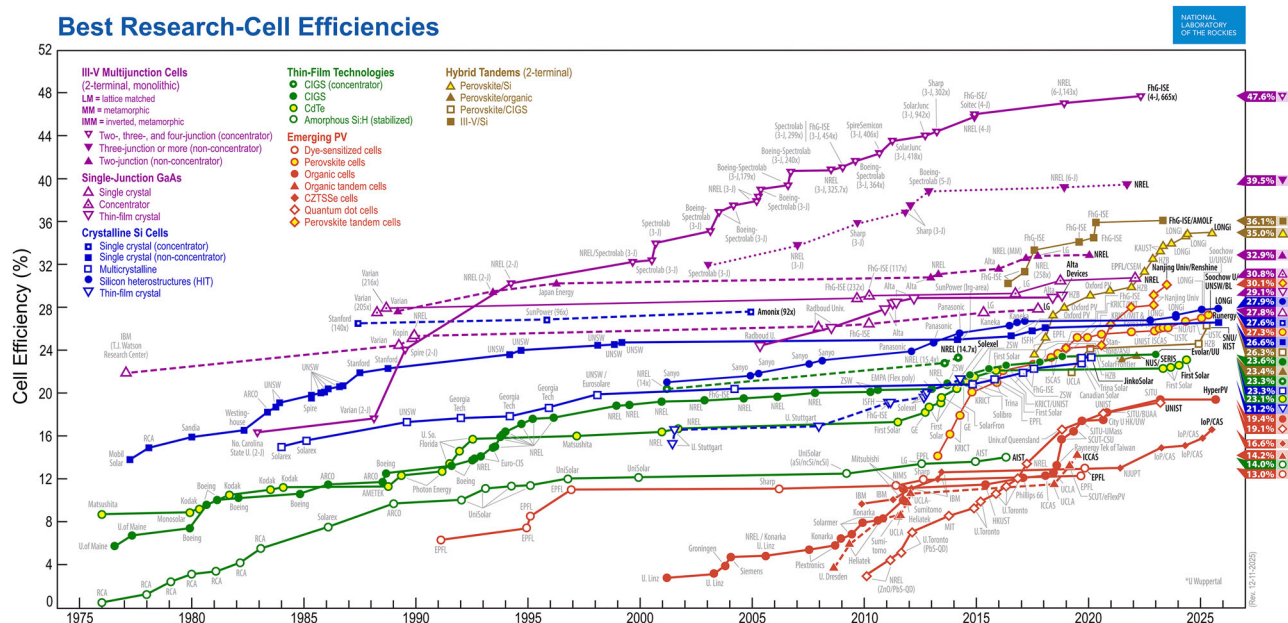


Fig. 2 | Evolution of photovoltaic research-cell efficiencies. Evolution of the certified power conversion efficiencies of representative photovoltaic technologies. Adapted from the National Renewable Energy Laboratory Best Research-Cell Efficiencies chart (<https://www.nrel.gov/pv/cell-efficiency>).

reduce defect density, and enhance device stability. These advances open new avenues for the development of more reliable and scalable Sn-based perovskite solar cells. Beyond solvent engineering, additive strategies have been widely employed to suppress defect formation and regulate crystallization in Sn-based perovskites. For example, tin-based additives such as $\text{Sn}(\text{SCN})_2$ effectively passivate Sn vacancies and suppress Sn^{4+} formation, thereby reducing nonradiative recombination and enhancing device performance⁶⁷. For example, Kang et al. introduced a strong coordinating reducing agent 4-phenylthiosemicarbazide (4PTSC), which optimized the morphology of Sn-HP films and reduced defects⁶⁸. The strong coordination between 4PTSC and SnI_2 enhances solution stability and delays perovskite crystallization, promoting grain growth. It also passivates Sn^{2+} and halide ions, suppresses the formation of SnI_4 , and reduces defects. The hydrophobic nature of 4PTSC prevents the oxidation of Sn^{2+} , thereby enhancing stability. As a result, the device open-circuit voltage increased from 0.78 V to 0.94 V, achieving a PCE of 12.22% (certified 11.70%), while maintaining excellent stability over 1200 h. Wei et al. synthesized two pyridine-substituted fulleropyrrolidine (PPF) derivatives with cis (CPPF) and trans (TPPF) configurations as precursor additives⁶⁹, as shown in Fig. 3a. The spatial configuration significantly influences their electronic distribution and interactions with perovskite components. Compared with CPPF, the spatially separated pyridyl groups in TPPF can effectively coordinate with perovskite colloids, slowing crystallization and forming films with improved orientation and higher density (Fig. 3b). TPPF remains stable at grain boundaries, optimizing energy-level alignment and suppressing Sn^{2+} oxidation. TPPF-based tin perovskite solar cells achieved a high efficiency of 15.38% (certified 15.14%) (Fig. 3f) and retained 99% and 93% of their initial efficiency after 3000 h of storage and 500 h of continuous illumination, respectively (Fig. 3g–h). While this example illustrates a typical strategy for defect passivation and crystallization control in Sn-based perovskites, similar approaches have been widely explored.

Furthermore, optimization of the charge transport layers is critical for enhancing the performance of Sn-based PSCs. Interface engineering plays a critical role in suppressing interfacial recombination and facilitating efficient charge extraction. For instance, the incorporation of functional interlayers or conjugated polymers into charge transport layers effectively reduces trap states and enhances device stability⁷⁰. Self-assembled monolayers (SAMs) have revolutionized the fabrication of PSCs^{71–75}, yet their application in Sn-based perovskite systems remains underexplored.

PEDOT is the preferred hole-transport layer in Sn-based PSCs^{76–79}, but it poses challenges for both performance and stability. [2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic Acid (MeO-2PACz) can be used to fabricate Sn-based PSCs⁸⁰, yet its performance consistently lags behind that of PEDOT. In recent studies, Musienko and co-workers designed a novel SAM-forming molecule named 2-[3,7-di(thiophen-3-yl)-10H-phenothiazin-10-yl]ethyl phosphonic acid (Th-2EPT)⁸¹, as shown in Fig. 4a–b. Th-2EPT optimizes coordination strength and enhances lattice compatibility of FASnI₃, facilitating the formation of high-quality buried interfaces and significantly suppressing nonradiative recombination, leading to a marked reduction in interfacial defects and an increase in carrier lifetime within the perovskite films. Figure 4c–e illustrates the calculated binding energies and interfacial configurations, revealing strong interactions between SAM molecules and the Sn-based perovskite surface that are critical for defect passivation and interface stabilization. As a result, the device achieved a high PCE of 8.2% (Fig. 4f–i). Qi et al. introduced an organic molecule⁸², that is (E)-2-(4',5'-bis(4-(bis(4-methoxyphenyl)amino)phenyl)-[2,2'-bithien]-5-yl)-1-cyanoethenylphosphonic acid, at the buried interface of inverted-structure Sn-based PSCs to form a uniform SAM layer⁸². This molecular layer promoted ideal energy-level alignment, effectively facilitated rapid hole extraction, and its superhydrophilic properties guided the ordered growth of perovskite crystals, significantly reducing defect density and nonradiative recombination losses. Ultimately, the fabricated inverted-structure small-area Sn-based PSCs achieved a record efficiency of 17.89% (certified 17.71%)⁸² and exhibited excellent stability—retaining over 95% of its initial efficiency after 1344 hours of storage in air and maintaining more than 94% efficiency after 1550 hours of continuous illumination. Moreover, a high efficiency of 14.40% was achieved on a 1 cm² large-area device, demonstrating the scalability and practical potential of this strategy. Although the efficiency still falls short of that of lead-based PSCs, it provides a new approach toward high-performance and stable lead-free perovskite devices.

While these results highlight the effectiveness of interfacial engineering in improving device performance, the underlying enhancements can be attributed to several dominant physical processes with well-defined causal relationships. Interfacial modification reduces defect density at the perovskite/transport-layer interface, suppressing trap-assisted nonradiative recombination and increasing quasi-Fermi level splitting, thereby enhancing the open-circuit voltage. Meanwhile, optimized energy-level alignment

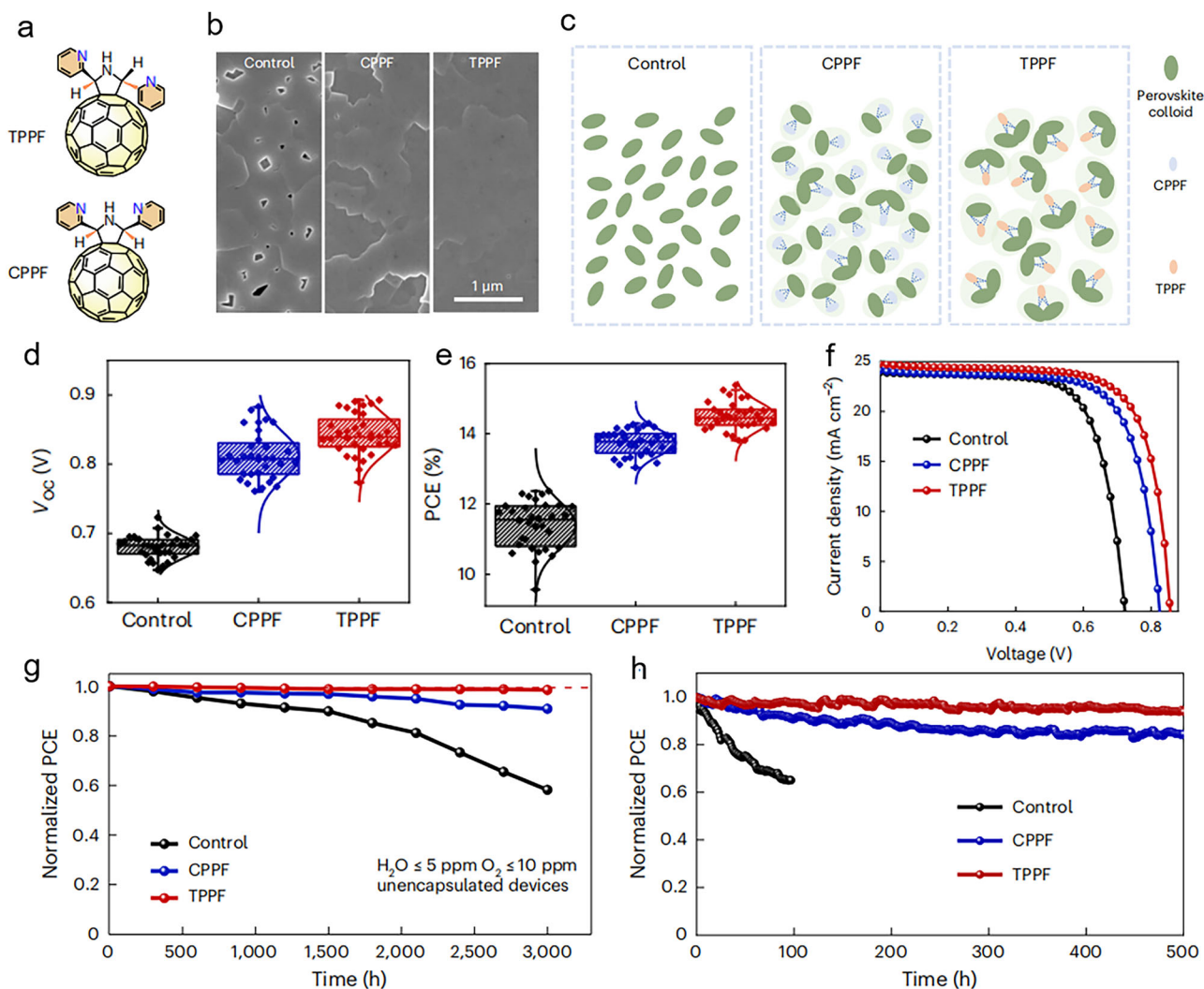


Fig. 3 | Effects of CPPF and TPPF additives on film quality and device performance. **a** Molecular structures of TPPF and CPPF. **b** SEM images of the control, CPPF- and TPPF-based perovskite films. **c** Schematic illustration of the colloids in the corresponding precursor solutions. **d, e** Statistical distributions of V_{OC} and PCE.

f $J-V$ curves of the best-performing TPSCs. **g, h** Stability of unencapsulated TPSCs under N_2 storage and continuous illumination. Adapted with permission from ref. 64. Copyright 2024 Springer Nature.

enables efficient charge extraction and minimizes interfacial carrier accumulation, thereby suppressing bimolecular recombination under operating conditions. Furthermore, stabilization of the interfacial chemical environment suppresses ion migration and defect evolution, both of which critically govern long-term device stability. These effects are closely coupled: reduced interfacial recombination prolongs carrier lifetime and promotes efficient charge extraction, while suppressed ion migration stabilizes the electronic structure and recombination dynamics during device operation. Therefore, the performance enhancement arising from interfacial regulation stems from the synergistic control of recombination kinetics, carrier transport, and interfacial stability, thereby establishing a clear causal link between interfacial design and device efficiency and stability. To enable a more direct comparison of device performance, representative experimental data of Sn-based perovskite solar cells under comparable conditions are summarized in Table 1. This table is positioned at the end of the Sn-based perovskite section, following the discussion of defect regulation, additive engineering, and interfacial optimization strategies, and prior to the introduction of Ge-based systems. To improve the rigor of performance comparison, reported PCE values should be interpreted together with measurement and validation conditions, including device area, scan direction, stabilized power output, EQE-integrated J_{SC} , hysteresis behavior, encapsulation condition, atmosphere, and operational stability protocol. Because these parameters

are not consistently reported across the literature, the values summarized in Table 1 should be regarded as representative literature benchmarks rather than a strict ranking of intrinsic material performance. As shown in Table 1, recent advances in Sn-based perovskite solar cells have led to efficiencies approaching ~18%, although notable limitations in open-circuit voltage and operational stability remain compared to Pb-based counterparts. Compared with Sn^{2+} , the smaller Ge^{2+} cation gives rise to Ge-based perovskites (e.g., $MAGeI_3$, $FAGeI_3$) possessing stronger Ge–I bonds and higher lattice energy, which collectively yield superior chemical stability. Although numerous theoretical studies have highlighted the potential of Ge-based perovskites in photovoltaic applications, experimental validation remains relatively limited^{83–85}. In 2015, Stoumpos et al. successfully synthesized $CsGeI_3$ as a prototype compound and prepared its analogues⁸⁶, including $MAGeI_3$, $FAGeI_3$, $CH_3C(NH_2)_2GeI_3$, $C(NH_2)_3GeI_3$, $(CH_3)_3NHGeI_3$, and $(CH_3)_2C(H)NH_3GeI_3$. They systematically characterized their structural features, electrical properties, and optical behaviors. However, their bandgaps are concentrated in the 1.6–2.8 eV range, generally larger than ideal. Besides, they are prone to oxidation to Ge^{4+} , resulting in reduced electrical activity and increased interfacial trap states. Consequently, Ge^{2+} is more frequently used for partial doping in Sn-based systems to enhance lattice rigidity, suppress Sn^{2+} oxidation, and tuning the bandgap to a more favorable range. Overall, through isovalent Sn–Ge substitution combined with rational

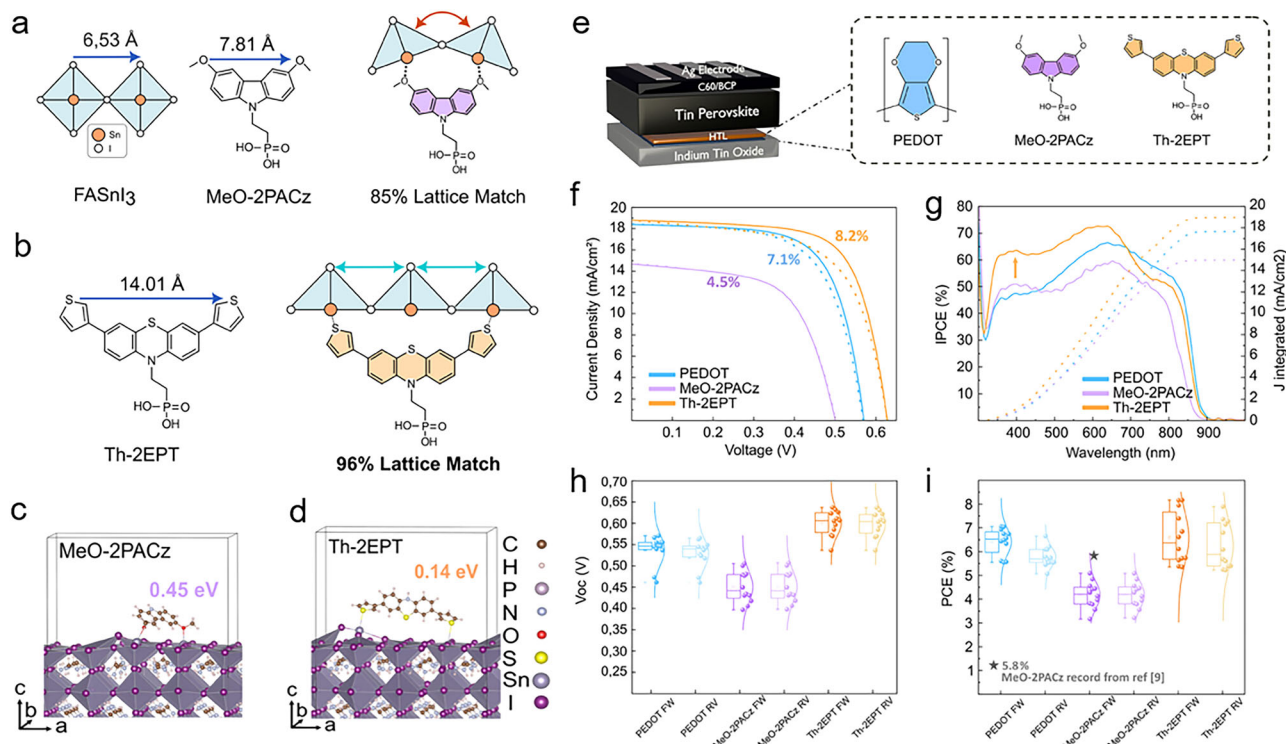


Fig. 4 | Interface engineering of lead-free perovskite solar cells. a Structures and characteristic dimensions of the FASnI₃ lattice and MeO-2PACz phosphonic acid, together with a schematic illustration of the lattice mismatch between MeO-2PACz and the FASnI₃ lattice. **b** Molecular structure of Th-2EPT, the S–S distance between the head groups, and the corresponding lattice matching with the FASnI₃ lattice. **c, d** Density functional theory (DFT)-calculated binding energies of the two molecules on a SnI₂-terminated FASnI₃ surface. **e** Schematic illustration of the lead-free

perovskite solar cell architecture and the molecular structures of three hole-selective materials. **f** J–V characteristics and **g** external quantum efficiency (EQE) spectra of representative devices employing different hole-selective layers. **h** Open-circuit voltage (V_{OC}) and **i** power conversion efficiency (PCE) statistics of the corresponding devices. Adapted with permission from ref. 75. Copyright 2025 Wiley-VCH.

chemical, structural, and interface engineering, it is possible to maintain strong light absorption while significantly enhancing material stability and bandgap tunability, providing a vital pathway for the development of high-performance lead-free PSCs.

Beyond Group IV elements, the other perovskite systems incorporating copper (Cu), manganese (Mn), and zinc (Zn) have gradually emerged as important research directions for lead-free PSCs^{87–92}. These materials, through partial or complete substitution of Pb²⁺, can form diverse structures such as zero-dimensional and low-dimensional, exhibiting enhanced structural stability and environmental tolerance. Mn²⁺- and Zn²⁺-doped systems can effectively tune the band structure and crystal stability, demonstrating potential for developing low-toxicity, highly stable photovoltaic materials^{93–95}. Their main advantages lie in low toxicity and structural tunability, allowing bandgap optimization through ionic composition and lattice parameter adjustments. This enhances spectral absorption and improves thermal, moisture, and photostability. However, these systems still suffer from low carrier mobility, high defect density, and relatively wide bandgaps, resulting in PCEs significantly lower than those of Pb-based PSCs. In addition, precise control over doping sites and concentrations is challenging, often leading to local structural distortions and energy level mismatches that hinder charge separation and transport. To overcome these bottlenecks, researchers have employed strategies such as lattice engineering, surface passivation, and doping control to enhance performance. Examples include co-doping with Cu/Mn or Cu/Zn to tune the bandgap and lattice strain, introducing organic cations or low-dimensional structures to reduce defect density^{96,97}, and utilizing ligand passivation and interface modification to suppress nonradiative recombination and improve carrier transport⁹⁸. Additionally, synthesis under inert atmospheres and optimized encapsulation further enhance environmental stability.

Overall, element substitution strategies not only preserve continuous band structures and high optoelectronic activity in lead-free systems but, through precise chemical control, interface engineering, and doping optimization, can effectively address the chemical instability and oxidation issues of Sn²⁺/Ge²⁺. This approach thus demonstrates the potential to achieve efficient charge generation and improved environmental compatibility. Currently, Sn-based lead-free perovskite solar cells have achieved a record power conversion efficiency exceeding 17.89% and have made significant progress in stability, highlighting their broad potential as a sustainable alternative to Pb-based systems. To provide a clearer overview of development trends in lead-free perovskite materials, Table 2 summarizes representative literature-reported results, including experimentally demonstrated devices and, where clearly indicated, theoretical or simulated predictions. It should be noted that simulated or DFT-predicted efficiencies are not directly comparable with experimentally measured PCEs, because they usually rely on idealized assumptions regarding film quality, defect density, interfacial recombination, and charge extraction. Therefore, Table 2 is intended to illustrate material-development trends rather than to establish a strict efficiency ranking. As shown in Table 2, Sn-based perovskites currently deliver the most competitive experimentally demonstrated photovoltaic performance, whereas other lead-free systems remain at earlier stages with comparatively lower verified device efficiencies. As shown in Table 2, Sn-based perovskites deliver the most competitive photovoltaic performance, whereas other lead-free systems remain at early stages with comparatively lower efficiencies. This comparison underscores the critical role of defect control and interface engineering in narrowing the performance gap with Pb-based perovskites.

Table 1 | Representative literature-reported performance parameters of lead-free perovskite solar cells

Material	Device structure	PCE (%)	V_{OC} (V)	J_{SC} (mA cm ⁻²)	J_{int} (mA cm ⁻²)	FF (%)	Ref.
CsSn _{0.96} Zn _{0.04} Br ₃	ITO/PEDOT:PSS/Perovskite/C ₆₀ /BCP/Ag	2.59	0.35	13.99	~15.00	54.00	139
WBG Sn perovskite	ITO/PEDOT:PSS/Perovskite/C ₆₀ /BCP/Ag	11.52	0.87	16.25	15.94	81.32	140
FASn-based PSC	ITO/ETL/Perovskite/HTL/Metal	16.06	9.85	25.79	24.71	73.26	70
FASnI ₃	ITO/PEDOT:PSS/FASnI ₃ /ETL/Al	14.30	0.92	20.40	20.30	76.70	141
FASnI ₃	ITO/PEDOT:PSS/FASnI ₃ /C ₆₀ /BCP/Al	12.22	0.70	24.11	~24.00	72.00	142
FASnI ₃	ITO/PEDOT:PSS/FASnI ₃ /ICBA/BCP/Ag	15.33	0.84	24.72	~23.90	0.736	143
FASnI ₃ (FPEABr)	ITO/PEDOT:FASnI ₃ /ICBA/Ag	15.70	0.97	21.70	/	74.10	144
2D/3D Sn PSC	ITO/NiO _x /Perovskite/PCBM/Ag	15.02	/	23.20	22.72	/	145
2D–3D Sn PSC	ITO/NiO _x /2D–3D/PCBM/Ag	9.41	0.61	22.00	20.80	70.10	146
FASnI ₃	FTO/Perovskite/HTL/Metal (module)	5.70	4.85	16.04	/	/	147
CsGel ₃	FTO/TiO ₂ /CsGel ₃ /MoS ₂ /Ni	13.57	/	/	/	/	148

Reported values should be compared cautiously because device area, measurement protocol, encapsulation condition, and stability-testing conditions vary among studies.

Table 2 | Development trends of lead-free perovskite materials

Material	Year	PCE (%)	V_{oc} (V)	Bandgap (eV)	Strategy	Ref.
CH ₃ NH ₃ SnI ₃	2014	/	/	~1.30	SnF ₂ doping/carrier transport	149
(PEA) ₂ SnI ₄ /FASnI ₃ hierarchy structure	2018	9.41	0.61	~1.20–1.40	2D–3D structure engineering	146
FASnI ₃	2020	~13.00	0.42	~1.30	band alignment engineering	150
FASnI ₃	2022	12.22	0.70	~1.30	surface passivation	142
FASnI ₃	2022	14.00	0.92	~1.30	crystallization control	141
FASnI ₃	2023	5.70	4.85	~1.30	scalable blade coating	147
FASnI ₃	2024	15.33	0.84	~1.30	energy-level tuning	143
FASnI ₃ + FPEABr	2024	15.70	0.97	1.42	interfacial dipole engineering	144
CsGel ₃	2025	13.57	/	~1.60	interface engineering (rGO)	148
CsSn _{1-x} Ge _x I ₃	2025	23.55 (simulated)	/	tunable	Alloy/DFT optimization	151
2D/3D Sn PSC	2025	15.02	/	~1.30	dimension–crystallization engineering	145

Aliovalent substitution of Pb

Beyond divalent cation substitution, trivalent pnictogen elements such as Bi³⁺ and Sb³⁺ have been widely explored as promising alternatives to Pb²⁺ in lead-free metal halide perovskites. Due to the charge mismatch between Pb²⁺ and Bi³⁺/Sb³⁺, direct one-to-one substitution is not feasible; instead, charge neutrality is achieved via vacancy-ordered or low-dimensional structures, such as A₃Bi₂X₉ and A₃Sb₂X₉ (A = MA, FA, Cs; X = Cl, Br, I). These materials exhibit enhanced chemical stability, reduced ion migration, and intrinsically lower toxicity, rendering them promising candidates for environmentally benign photovoltaic applications. From an optoelectronic perspective, Bi- and Sb-based perovskites exhibit wide bandgaps (>1.8 eV), indirect or quasi-indirect transitions, and strong exciton binding, collectively limiting carrier transport and photovoltaic efficiency in conventional single-junction solar cells. In addition, the incorporation of pnictogen cations can modify the local coordination environment and defect chemistry, thereby influencing trap-state formation, carrier recombination dynamics, and interfacial energy alignment. As a result, their power conversion efficiencies remain markedly lower than those of Sn-based systems. Nevertheless, these intrinsic characteristics define a unique application niche, particularly in indoor photovoltaics (IPV), where Bi- and Sb-based metal halide perovskites exhibit considerable potential. Indoor light sources, such as light-emitting diodes and fluorescent lamps, typically feature low intensity and narrow spectral distributions. The relatively wide bandgaps of Bi- and Sb-based materials align well with these spectra, enabling efficient photon utilization under low illumination and minimizing thermalization losses. Under indoor illumination conditions, photovoltaic materials are more sensitive to trap-assisted nonradiative recombination because of the

much lower carrier generation rate compared with one-sun illumination. Therefore, materials with low defect densities, stable interfaces, and suppressed ion migration are particularly important for efficient indoor photovoltaics. In this respect, Bi- and Sb-based perovskites exhibit potential advantages owing to their enhanced environmental stability and defect-tolerant characteristics⁹⁹. Meanwhile, their low toxicity and superior environmental stability make them highly desirable for indoor applications, where device safety and long-term reliability are critical. These wide-bandgap materials are promising candidates for top-cell absorbers in tandem and multifunctional photovoltaic architectures. Their large bandgaps enable efficient utilization of high-energy photons, while lower-bandgap materials (e.g., Sn-based perovskites or silicon) act as bottom cells to harvest the remaining spectrum. However, the use of Bi- and Sb-based materials in tandem devices remains at an early stage, owing to their limited absorption coefficients, low carrier mobility, and suboptimal energy-level alignment.

To address these limitations, a range of strategies, including dimensionality engineering, defect passivation, heterostructure construction, and compositional tuning, have been explored. While these approaches have yielded incremental improvements, achieving a balance among optical absorption, carrier transport, and long-term stability remains a key challenge. Therefore, Bi- and Sb-based perovskites should not be considered direct high-efficiency replacements for Pb-based systems, but rather complementary materials with distinct advantages in low toxicity, stability, and application-specific photovoltaic technologies. Beyond vacancy-ordered or low-dimensional structures, heterovalent substitution provides an alternative strategy for incorporating trivalent cations such as Bi³⁺ and Sb³⁺ into lead-free perovskite frameworks. In this approach, charge neutrality is

achieved through the simultaneous introduction of a monovalent cation, resulting in double-perovskite structures. Compared with low-dimensional Bi- and Sb-based compounds, double perovskites retain a three-dimensional octahedral framework, enabling more efficient carrier transport and improved device integration. This strategy thus offers an effective route to combine the enhanced stability and low toxicity of Bi- and Sb-based systems with improved electronic connectivity and structural robustness.

In a typical ABX_3 halide perovskite, the B-site is occupied by a divalent metal cation such as Pb^{2+} . Aliovalent substitution replaces two Pb^{2+} ions with one monovalent cation (M^+) and one trivalent cation (M^{3+}), thereby maintaining charge neutrality and forming vacancy-free halide double perovskites with the general formula $A_2MM^+X_6$ ^{100–102}. It should be emphasized that the double perovskites discussed in this review refer specifically to halide double perovskites, which are fundamentally distinct from oxide perovskites. Oxide perovskites are based on metal-oxygen frameworks and are often associated with strong metal-oxygen bonding, correlated-electron behavior, ferroelectricity, and catalytic functionality. In contrast, halide double perovskites consist of softer corner-sharing metal-halide octahedra and generally exhibit different defect chemistry, processing routes, band-edge orbital compositions, and optoelectronic application directions. The ordered arrangement of monovalent and trivalent cations at the B site forms a rock-salt-like superlattice, which can enhance structural rigidity and oxidation resistance but also substantially alters the electronic structure relative to Pb halide perovskites. For representative systems such as $Cs_2AgBiBr_6$ and $Cs_2AgInCl_6$, the band-edge states are determined by the hybridization among Ag/Cu d orbitals, Bi/Sb/In p orbitals, and halide p orbitals^{103,104}. In $Cs_2AgBiBr_6$, for example, the valence-band edge contains significant Ag 4d–Br 4p antibonding character, whereas the conduction-band edge is mainly associated with Bi 6p orbitals. The ordered heterovalent B-site arrangement can reduce band dispersion and place the valence-band maximum and conduction-band minimum at different k-points, leading to indirect bandgaps and weak near-edge optical absorption. In addition, parity-forbidden or symmetry-suppressed transitions may further reduce absorption strength even when the nominal bandgap appears suitable. Therefore, the photovoltaic limitations of many halide double perovskites originate not only from wide bandgaps but also from their intrinsic orbital and symmetry characteristics. Halide or metal substitution can tune the bandgap to some extent; however, this strategy should be considered critically rather than treated as a universally applicable route for photovoltaic optimization. Most experimentally realized three-dimensional halide double perovskites are based on Cl- and Br- systems, whereas iodide-rich analogues remain difficult to stabilize because of unfavorable lattice energetics, enhanced structural distortion, phase instability, and possible defect formation¹⁰⁵. Although approaches such as metal alloying, symmetry breaking, cation disorder, low-dimensional structural design, and defect engineering have been explored to improve optical absorption and carrier transport, achieving a direct or strongly allowed bandgap while simultaneously preserving phase stability and defect tolerance remains a major unresolved challenge^{105–108}.

The performance of double PSCs currently lags behind, with PCEs typically under 5%. Nevertheless, continuous improvements in crystal quality, interface engineering, and electronic structure provide a clear roadmap for rapid performance gains. Zhu et al. introduced fused-ring electron acceptor (FREA) molecules to passivate defects in $Cs_2AgBiBr_6$ -based double PSCs¹⁰⁹, as shown in Fig. 5a. The $C\equiv N$ and $N=C-S$ functional groups in FREA molecules strongly interact with exposed Ag atoms on the $Cs_2AgBiBr_6$ surface, effectively reducing surface trap density and significantly suppressing nonradiative recombination. This resulted in a significant increase in V_{OC} , from 1.079 V to 1.278 V, and achieved a maximum device efficiency of 3.31% (Fig. 5b). Furthermore, the passivated devices exhibited excellent long-term stability, retaining 98.5% and 97.2% of their initial efficiency after 300 hours under continuous illumination and heating at 85 °C, respectively (Fig. 5c–d). Sui and colleagues employed a hydrogenation treatment to tune the bandgap of $Cs_2AgBiBr_6$ films¹¹⁰, adjusting it from 2.18 eV to 1.64 eV, as shown in Fig. 4f. Meanwhile, the hydrogenated

$Cs_2AgBiBr_6$ PSCs achieved a maximum PCE of 6.37% and exhibited excellent environmental stability (Fig. 5h). Interstitial hydrogen doping not only tunes the valence and conduction-band levels of $Cs_2AgBiBr_6$ but also optimizes carrier mobility and lifetime. These studies demonstrate that defect passivation and band-structure modulation can improve double-perovskite devices; however, they also highlight that halide double perovskites remain more valuable as stable lead-free semiconductor platforms than as direct high-efficiency replacements for Pb-based photovoltaic absorbers.

Although $Cs_2AgBiBr_6$ -based double PSCs have made significant progress in stability and defect passivation, their relatively wide indirect bandgap (approximately 1.95–2.2 eV) still limits light absorption, thereby constraining the PCE of the devices. The indirect bandgap in most double perovskites arises from their unique electronic structure, where the valence-band maximum and conduction band minimum originate from different orbital contributions (e.g., Ag 4d, Bi 6p, and halide p orbitals), leading to momentum mismatch in optical transitions. In addition, the ordered arrangement of heterovalent cations (M^+ and M^{3+}) induces symmetry-driven band splitting and reduces orbital overlap, thereby suppressing direct transitions. As a result, the absorption coefficient and radiative recombination rate are significantly reduced, thereby contributing to the discrepancy between theoretically predicted and experimentally achieved device efficiencies. To overcome this limitation, researchers have begun exploring lead-free double-perovskite materials with tunable electronic structures^{111,112}. By tailoring the composition through partial-to-complete substitution of Bi^{3+} with In^{3+} , combined with halide composition (Cl/Br) tuning, the bandgap and carrier mobility of $Cs_2AgInCl_{6-x}Br_x$ can be strategically tuned, thus enhancing light absorption and carrier transport properties while preserving the inherent stability and low toxicity of the double-perovskite structure^{113,114}. However, such bandgap-engineering strategies often entail a trade-off between optical performance and material stability. For example, halide substitution or metal alloying can reduce the bandgap and enhance light absorption, yet may introduce structural disorder, increased defect density, or phase instability. Therefore, achieving a balance between bandgap optimization and long-term stability remains a key challenge in double-perovskite solar cells. This transition reflects a strategic shift from solely prioritizing stability to balancing both efficiency and long-term reliability, providing an effective route to enhanced light absorption and carrier extraction in lead-free photovoltaic materials. Tripathi and co-workers systematically investigated the structural, electronic, and optical properties of $Cs_2AgInCl_{6-x}Br_x$ ($x = 0–6$) via DFT and evaluated their photovoltaic performance using SCAPS-1D simulations (Fig. 6a–b)¹¹⁵. The results suggest that increasing Br content induces lattice expansion and bandgap narrowing, while the conduction band minimum and valence-band maximum become more dispersed, indicating improved carrier mobility. $Cs_2AgInClBr_5$ is predicted to exhibit favorable optical properties, including strong visible-light absorption and reduced reflection. Device simulations further suggest that this composition exhibits improved carrier transport and a direct bandgap of ~1.75 eV (Fig. 6c), yielding a simulated PCE of up to 21.95% (Fig. 6d–g). However, it should be noted that these results are based on theoretical calculations and idealized device models. In practice, achieving such high performance remains challenging owing to the intrinsic limitations of double perovskites, including indirect bandgaps, defect states, and limited carrier transport in experimentally accessible systems.

Although material modification is a core approach for improving double-perovskite photovoltaics, their intrinsic properties, such as indirect bandgaps and unfavorable carrier dynamics, impose performance ceilings that cannot be overcome by compositional adjustments alone. Consequently, the research paradigm is shifting from material modification to device architecture. Key strategies—including interface engineering, optimized charge transport layers, and precise energy-level alignment—are being leveraged to enhance light harvesting, carrier extraction, and overall device performance. Alanazi et al. conducted design and numerical simulations of $Cs_2AgBiBr_6$ double PSCs using the SCAPS-1D device simulator¹¹⁶.

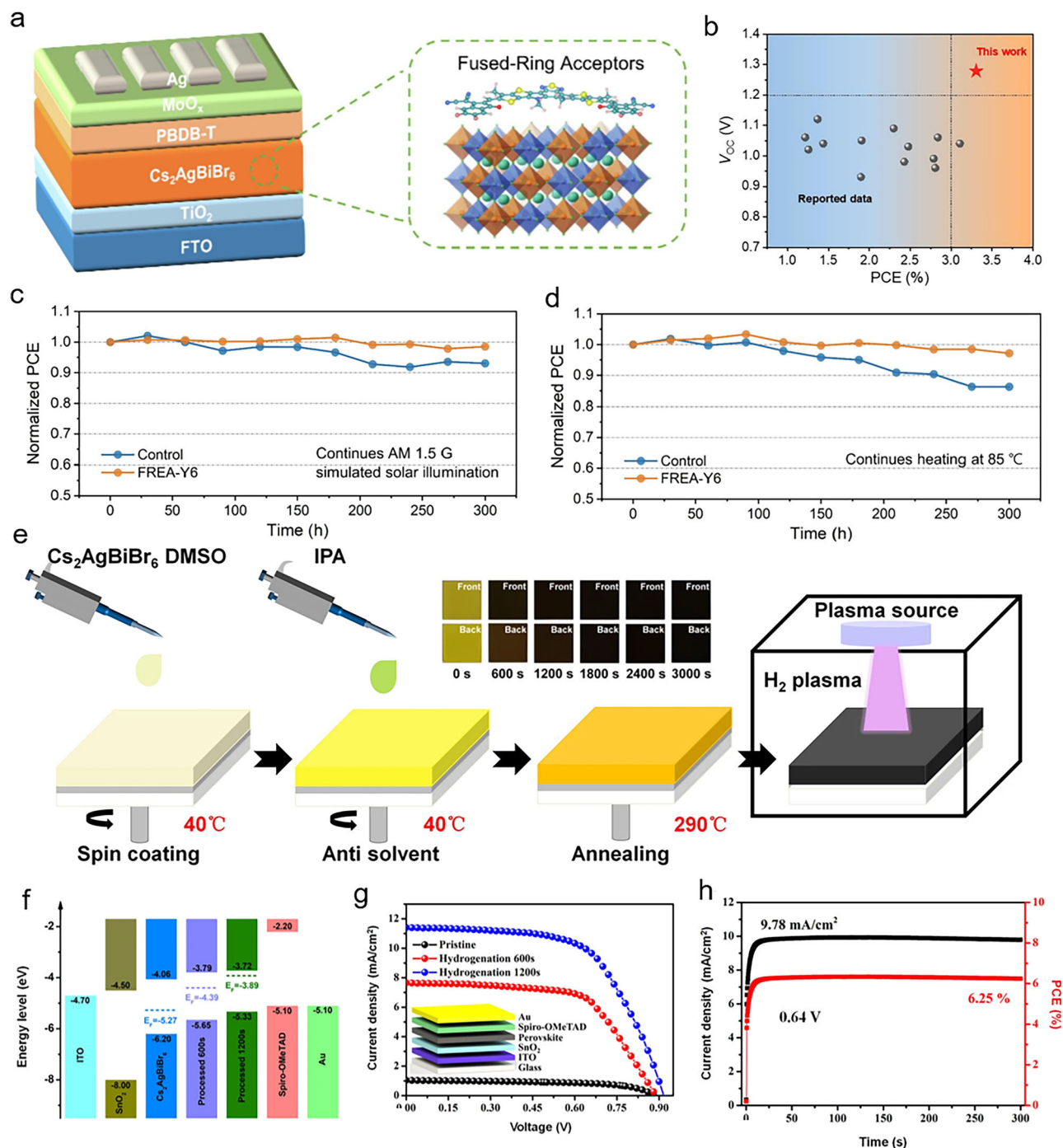


Fig. 5 | Summary of additive and interfacial engineering strategies for $\text{Cs}_2\text{AgBiBr}_6$ perovskite solar cells. **a** Reported device architecture of $\text{Cs}_2\text{AgBiBr}_6$ perovskite solar cells and the molecular structure of FREA. **b** Comparison of device performance (PCE and V_{oc}) of $\text{Cs}_2\text{AgBiBr}_6$ perovskite solar cells with previously reported values. **c, d** Operational stability of $\text{Cs}_2\text{AgBiBr}_6$ perovskite solar cells with and without FREA-Y6 under continuous illumination and thermal stress, respectively. **e** Schematic illustration of reported hydrogen plasma treatment for

$\text{Cs}_2\text{AgBiBr}_6$ perovskite films, together with representative evolution of film appearance with treatment time. **f** Energy-level alignment of functional layers in hydrogen-treated $\text{Cs}_2\text{AgBiBr}_6$ perovskite solar cells. **g** $J-V$ characteristics of devices treated with different hydrogenation times (0, 600, and 1200 s). **h** Steady-state photocurrent of the optimized device near its maximum power point. Adapted with permission from ref. 94. Copyright 2022 Elsevier; and ref. 95. Copyright 2022 Springer Nature.

The initial calibration device was based on experimental studies, in which the $\text{Cs}_2\text{AgBiBr}_6$ absorber layer was hydrogenated to achieve the lowest reported bandgap to date ($E_g = 1.64 \text{ eV}$). The device with the structure ITO/ SnO_2 / $\text{Cs}_2\text{AgBiBr}_6$ /Spiro-OMeTAD/Au, achieved a maximum PCE of 6.58%. Their systematic study identified the electron transport layer (ETL)/ $\text{Cs}_2\text{AgBiBr}_6$ conduction-band offset as a decisive factor for PCE, also highlighting the significant influence of absorber layer thickness. By

employing a double ETL structure of $\text{SnO}_2/\text{ZnO}_{1-x}\text{S}_x$ and optimizing absorber layer thickness, the PCE was increased to 14.23%. Shen and colleagues demonstrated that introducing Ag nanoparticles on the surface of $\text{Cs}_2\text{AgBiBr}_6$ double perovskites can significantly enhance device optoelectronic performance¹⁷. The Ag nanoparticles enhance the short-circuit current density (J_{sc}) through far-field light scattering and near-field plasmonic effects, while increasing the Fermi level difference at the TiO_2 /

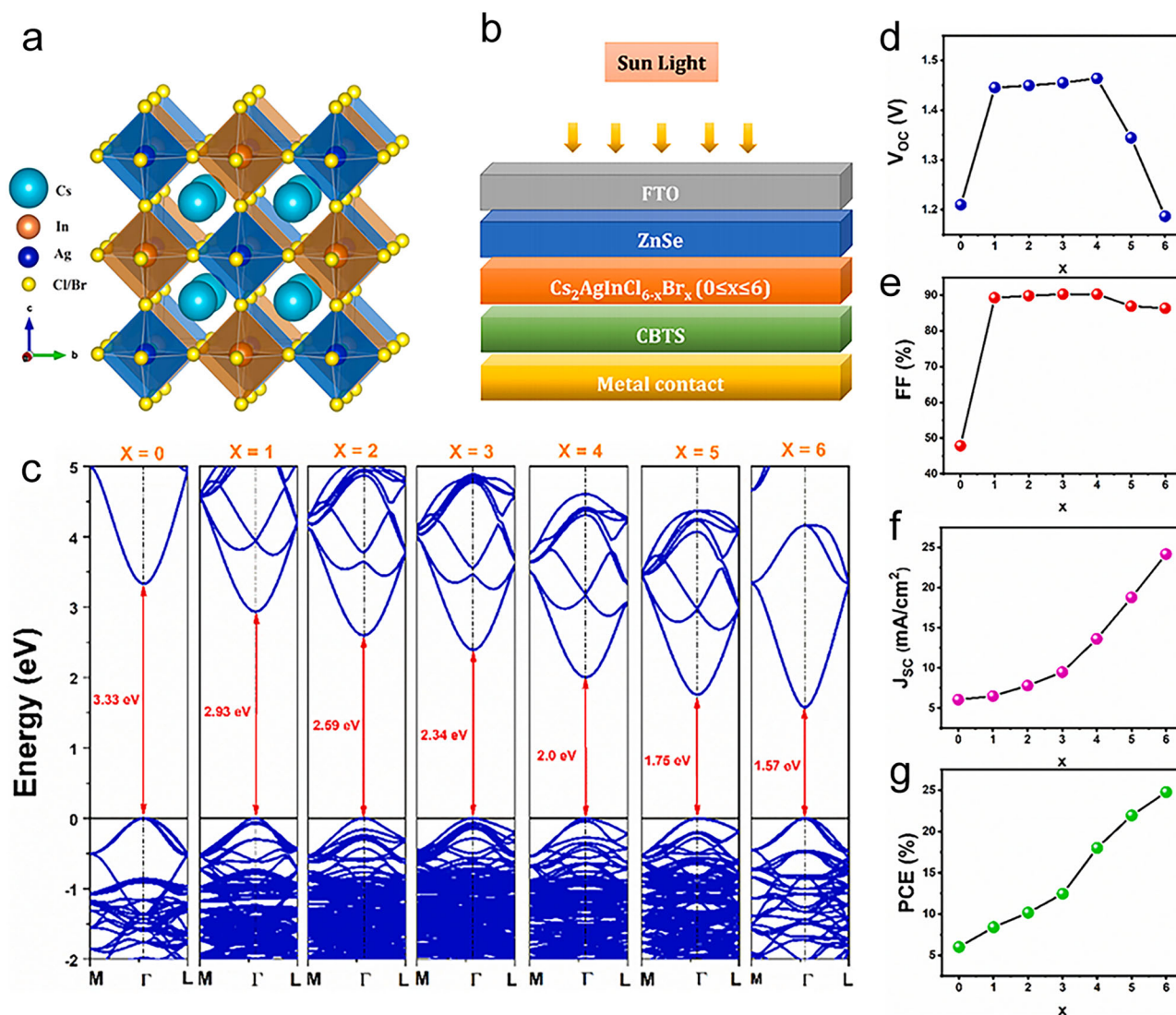


Fig. 6 | Effect of $\text{Cs}_2\text{AgInCl}_{6-x}\text{Br}_x$ composition on crystal structure and solar cell performance. a The crystal structure of $\text{Cs}_2\text{AgInCl}_{6-x}\text{Br}_x$ ($x = 0, 1, 2, 3, 4, 5$ and 6) DPs and **b** device configuration of the proposed PSCs. **c** The calculated electronic

band structure. **d–g** Influence of the absorber layer $\text{Cs}_2\text{AgInCl}_{6-x}\text{Br}_x$ on photovoltaic parameters such as V_{oc} , FF, J_{sc} , and PCE. Adapted with permission from ref. 150. Copyright 2024 Elsevier.

perovskite interface, thereby improving the open-circuit voltage. The optimized carbon-based, hole-transport layer (HTL)-free device achieved a PCE of 2.69%, representing a significant improvement over the control device. This strategy reflects a shift from solely material modification to a combined material–device optimization approach, offering a useful route to improve double-perovskite devices while recognizing that their high-efficiency photovoltaic application remains constrained by intrinsic absorption and carrier-transport limitations.

From an application perspective, halide double perovskites should be assessed more cautiously as photovoltaic absorbers. Although their low toxicity, high structural stability, and compositional diversity make them attractive lead-free semiconductors, their indirect bandgaps, weak absorption coefficients, and limited carrier transport impose intrinsic constraints on single-junction photovoltaic performance. Therefore, halide double perovskites should not be regarded simply as direct high-efficiency replacements for Pb-based perovskite absorbers. Instead, their strengths may be more fully exploited in stability-demanding or non-PV optoelectronic applications, such as photodetection, light emission, photocatalysis, radiation detection, and emerging quantum/optoelectronic technologies. For photovoltaic applications, their near-term relevance may lie in niche devices, indoor or semitransparent photovoltaics, tandem auxiliary layers, or

model systems for understanding defect-tolerant lead-free semiconductors.

In summary, halide double perovskites provide an important platform for exploring lead-free metal-halide semiconductors with improved environmental compatibility, chemical stability, and structural diversity. However, their indirect bandgaps, symmetry-constrained optical transitions, and limited carrier transport currently impose significant barriers to high-efficiency single-junction photovoltaics. Future progress requires not only empirical compositional screening but also rational band-edge orbital engineering, symmetry control, defect-tolerant cation combinations, improved film formation, and device architectures that compensate for weak absorption.

Novel environmentally friendly lead-free perovskites

As photovoltaic materials advance toward higher efficiency, environmental friendliness, and reduced toxicity, researchers have increasingly turned their attention to novel environmentally friendly perovskites. Currently, two classes of novel perovskites have attracted significant attention. One is lead-free sulfide perovskites, where chalcogen elements replace X-site halide, achieving broad-spectrum absorption and favorable carrier transport property¹¹⁸. The other is metal-free organic perovskites, in which the B-site

of the ABX_3 structure is occupied by NH_4^+ and paired with various organic divalent cations and halide anions, forming fully metal-free light-absorbing materials that combine structural flexibility with potential ferroelectricity^{119,120}. These two classes of materials not only reduce environmental and health risks but also exhibit excellent bandgap tunability and optical properties.

Sulfide perovskites arise from the partial or complete substitution of halide anions with sulfur, introducing a markedly different bonding environment compared with conventional lead-halide counterparts. This anion substitution imparts distinct optical and electronic advantages, reflected in the broad tunability of their electronic structures, which are highly sensitive to the choice of B-site cation. Small-bandgap sulfide perovskites, including Sn-S and Ge-S derivatives, exhibit direct bandgaps that span the visible to near-infrared region, enabling broad spectral coverage and efficient photogeneration. Their intrinsically low carrier effective masses further support rapid transport and efficient charge extraction. By contrast, wide-bandgap chalcogenide perovskites, such as $BaZrS_3$ and $LaYS_3$, feature bandgaps exceeding ~ 1.8 eV. Although their absorption edges lie within the visible spectrum, the strongly covalent B-S framework confers exceptional thermal, moisture and photochemical robustness—qualities that directly address the persistent instability challenges of lead-halide systems. These materials can be processed into dense and uniform thin films via solution deposition or chemical vapor routes. Their direct-bandgap character enables efficient absorption in the visible regime and yields respectable photocurrent densities, positioning them as promising candidates for durable photovoltaics and tandem configurations. Among various chalcogenide perovskites, $BaZrS_3$ has emerged as one of the most representative and extensively studied systems because of its suitable bandgap, strong optical absorption, and remarkable thermal and moisture stability^{121,122}. Through interface engineering and surface defect passivation, trap-state density can be significantly reduced, enhancing carrier extraction efficiency^{121,123,124}. $BaZrS_3$ sulfide perovskite is a promising high-efficiency light absorber with excellent optical absorption and carrier transport properties; however, experimental studies remain limited, and its actual photovoltaic performance requires further validation. Marasamy et al. performed a theoretical study to evaluate the potential of $BaZrS_3$ for photovoltaic applications¹²⁵. Their results suggest that a device with the structure FTO/ ZrS_2 / $BaZrS_3$ /SnS/Pt can achieve a maximum PCE of 32.58% following Ti doping optimization, owing to reduced interfacial barriers, enhanced carrier transport, and extended near-infrared absorption. Notably, such performance is based on theoretical calculations under idealized conditions, and its translation into experimentally achievable device performance remains challenging. This study thus provides a theoretical foundation for the design of high-efficiency $BaZrS_3$ -based photovoltaic devices.

Furthermore, sulfide perovskites can be integrated with narrow-bandgap PSCs or silicon solar cells to form tandem or heterojunction structures, enabling full-spectrum light absorption and further enhancing overall device performance^{126–128}. This provides a new pathway for the development of high-efficiency and environmentally friendly photovoltaic devices. In recent years, studies have also explored strategies such as alloying, elemental doping, and low-dimensional structure design to modulate band structure and carrier dynamics, aiming to further enhance photovoltaic conversion efficiency^{129,130}. Although the photovoltaic efficiency of laboratory-prepared sulfide perovskite solar cells is still lower than that of lead-based perovskites, their low toxicity, direct bandgap characteristics, strong light absorption, and excellent stability make them a promising research focus and a key direction for next-generation high-performance, environmentally friendly photovoltaic devices^{131,132}. In the future, with continuous advances in material design and device engineering, sulfide perovskites are expected to achieve efficient, stable, and sustainable energy conversion in practical photovoltaic applications.

Metal-free organic perovskites (MFOPs) are a class of perovskite derivatives in which traditional metal cations are completely replaced by organic cations. Their typical chemical formula is $A^{2+}(NH_4^+)X_3$, where A^{2+} represents a divalent organic cation (such as 1,4-diazabicyclo[2.2.2]octane

(DABCO) derivatives or N-methyl-1,4-diazabicyclo[2.2.2]octan-1-ium ($MDABCO^{2+}$)), and X is a halide or halide-like anion. These materials maintain the perovskite-type structure through intermolecular hydrogen bonding and electrostatic interactions, avoiding the environmental and toxicity issues associated with heavy metals such as lead or tin. Since their energy bands are mainly composed of organic molecular orbitals and halide orbitals¹³³, MFOPs typically exhibit tunable bandgaps (1.7–4.0 eV) and excellent optical transparency. Some systems also display intrinsic ferroelectricity, which can generate spontaneous polarization fields in devices, promoting charge carrier separation and transport. Although their carrier mobility is relatively low, the combination of low trap-state density and tunable polarization endows MFOPs with unique optoelectronic modulation potential¹³⁴. In PSCs, the application of MFOPs is evolving from auxiliary layers toward active absorbers. Currently, they are primarily employed for nucleation control and interface passivation, improving the crystallinity and stability of Pb- or Sn-based perovskite films and thereby enhancing device V_{OC} and operational lifetime. For example, MDABCO-based metal-free perovskites have been explored as light absorbers owing to their tunable bandgaps and intrinsic ferroelectricity, demonstrating their potential for photovoltaic applications. Theoretically, the bandgap of MFOPs can be tuned to 1.6–1.8 eV through cation conjugation engineering or I^- incorporation¹³⁵. However, their practical photovoltaic performance remains limited. Nevertheless, their wide bandgaps make them promising candidates for top-cell applications in tandem solar cells, although further improvements in carrier transport and film quality are still needed.

While elemental abundance and toxicity are frequently used as primary indicators for evaluating lead-free perovskite materials, they are insufficient on their own to provide a comprehensive assessment of sustainability. Recent life-cycle assessment (LCA) studies on perovskite photovoltaics have shown that environmental impacts depend not only on absorber toxicity, but also on precursor production, solvent use, electrode and transport-layer materials, fabrication temperature, module lifetime, encapsulation, recycling efficiency, and end-of-life management. Therefore, replacing Pb with less toxic elements does not automatically guarantee a lower overall environmental burden. For example, Pb-based PSCs raise concerns related to potential lead leakage and toxicity under damage, fire, or improper disposal scenarios, whereas some lead-free systems may rely on scarce or resource-critical elements, exhibit lower device efficiencies, or suffer from shorter operational lifetimes, thereby increasing the environmental impact per generated kilowatt-hour. In particular, Ag-, In-, Bi-, and Sn-containing lead-free perovskites should be evaluated not only in terms of acute toxicity, but also in terms of resource availability, mining and purification impacts, chemical stability, recyclability, and realistic device lifetime. A meaningful sustainability comparison between Pb-based and lead-free PSCs therefore requires cradle-to-grave LCA covering raw material extraction, material synthesis, device fabrication, operation, failure/leakage scenarios, recycling, and final disposal. Key indicators should include carbon footprint, cumulative energy demand, energy payback time, human toxicity potential, ecotoxicity potential, resource criticality, recycling yield, and environmental impact per unit electricity generated. Accordingly, future lead-free perovskite research should integrate LCA-guided material selection, low-toxicity solvents, low-temperature processing, durable encapsulation, and closed-loop recovery strategies from the earliest stage of device design^{136–138}.

Outlook

The development of lead-free perovskite solar cells is shifting beyond simple Pb substitution toward an integrated optimization of efficiency, stability, and sustainability. Among various strategies, Sn-based perovskites remain the most promising candidates for high-performance lead-free photovoltaics, with reported small-area efficiencies approaching 18% and certified values above 17% in selected studies. However, these performance values should be interpreted together with device area, measurement protocol, stabilized output, EQE validation, and operational stability, because reproducibility and scalability remain unresolved challenges. However, their performance is fundamentally limited by the coexistence of extrinsic Sn^{2+}

oxidation and intrinsic vacancy-induced self-doping, resulting in instability and voltage loss. Future progress will require decoupling these effects through chemical stabilization, defect control, and interface engineering, alongside improved film uniformity and suppressed nonradiative recombination.

In contrast, aliovalent double perovskites and other emerging systems offer enhanced chemical stability and reduced toxicity, yet suffer from intrinsic limitations, including indirect bandgaps, low carrier mobility, and challenges in film formation. Addressing these issues requires precise control over lattice symmetry, orbital hybridization, and defect states, together with improved processing strategies and device architectures. Emerging materials, including chalcogenide and metal-free organic perovskites, broaden the material landscape but remain at an early stage, with key challenges in synthesis, carrier transport, and device integration. Interdisciplinary approaches, including high-throughput computation, machine learning, and in situ characterization, are expected to accelerate material discovery and provide deeper insight into degradation mechanisms.

From an application perspective, lead-free perovskites hold strong promise for emerging technologies, including indoor photovoltaics, tandem architectures, and flexible or wearable devices, where their tunable bandgaps and low toxicity offer distinct advantages. However, challenges in band alignment, interfacial recombination, and long-term operational stability must be overcome to enable practical implementation. Meanwhile, sustainability considerations should be integrated into future research through quantitative LCA metrics, including carbon footprint, cumulative energy demand, energy payback time, toxicity potential, resource criticality, recyclability, and end-of-life recovery efficiency. Such an approach is essential for fairly comparing Pb-based and lead-free systems, because lower absorber toxicity may be offset by lower efficiency, shorter lifetime, scarce-element use, or more energy-intensive processing. Overall, the advancement of lead-free perovskite photovoltaics relies on the synergistic optimization of material chemistry, device physics, and sustainable design, paving the way for safer and more scalable next-generation photovoltaic technologies.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

K.C. and G.L. conceived and designed the project. The paper was written by W.Z. and K.C. All authors commented on the data and on the final version of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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